Department of Master of Computer Application

Session 2025-2026

Progress Report of Mini-Project work – 1

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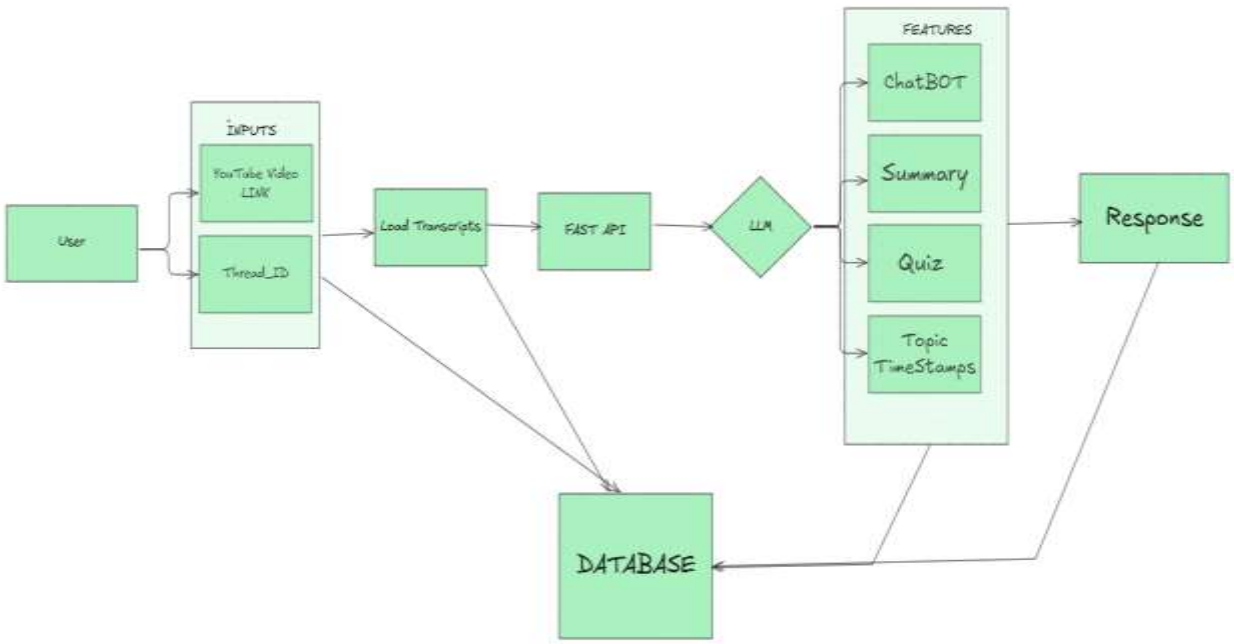
Precise Report of the Project work done during the above period (Brief)	
1	**Study the various paper related to research area** **1.Scope & trends** — Recent Blogs focus on combining Retrieval-Augmented Generation (RAG) with structured schemas to handle any written Content 2. **Transcription + retrieval** — Best results come from accurate transcripts followed by chunked retrieval (semantic search) rather than sending full transcripts to the LLM. **3.Privacy & cost** — Research highlights stored transcripts and using caching/checkpointing to reduce API tokens and latency.
2	**Find out the various Techniques / Algorithm** **1. RAG (Retrieval-Augmented Generation) —** Store transcript chunks in a vector DB; retrieve relevant chunks per query and feed to LLM for focused answers. **2. Structured Output / Schema Enforcement —** Ask LLM to output JSON with fields (summary, key_points[], timestamps[], quiz[]) for predictable parsing and DB storage. **3. Caching —** Cache embeddings, retrieval results, and common LLM outputs to avoid repeated token usage for the same video/thread. **4. Checkpointer / Conversation History Checkpointing —** Periodically checkpoint user-LLM sessions (store compressed context or summary) so long chats don't resend entire history. ○
3	**Analysis of existing algorithms and its weakness** **Non-RAG (naïve LLM on whole transcript) uses many tokens —** Cost and latency explode for long videos; prompts can exceed model limits. **No caching → repeated cost** — Systems that regenerate summaries/answers each time increase API bills and slow UX

4	**Analysis of existing algorithms with our system** **1. RAG reduces token usage —** TubeTalk uses chunked retrieval so only relevant chunks go to LLM, lowering cost and improving focus. **2. Structured schema yields reliable features** — By enforcing JSON schema (summary, quiz, timestamps), TubeTalk.ai stores and serves outputs cleanly to the UI and DB. **3. Caching & checkpointing improve speed & cost —** Cached embeddings and checkpoints let users re-open sessions or regenerate quizzes without new full LLM runs. **4. Timestamp extraction + overlap reduces boundary errors** — Our chunk overlap and timestamp alignment gives users accurate jump-to moments. **5. Fallback & logging increase robustness** — If LLM fails, TubeTalk falls back to stored cached outputs or lightweight extractive summaries and logs failures for debug and retry.
5	Publish the Introductory/ Survey paper on research topic: Yes/No
6	**Explore the flow of Project – TubeTalk.ai**  **Flow of TubeTalk.ai** **1. User Input & Authentication** o User provides YouTube video link and Thread_ID. o Request goes through frontend → backend (FastAPI). o Session is authenticated using JWT-based tokens for secure access. **2. Transcript Loading** o System fetches and loads YouTube transcripts via backend service. o Transcripts are stored temporarily and/or cached in the database for reuse.

	3. Processing via LLM o FastAPI sends transcripts to the LLM. o LLM generates multiple feature outputs: ▪ Chatbot (Q&A on video content) ▪ Summary (shortened version of content) ▪ Quiz (auto-generated questions) ▪ Topic Timestamps (segmented highlights) **4. Database Storage** o All outputs (transcripts, summaries, quizzes, timestamps) are encrypted and stored in the database. o Each entry linked with Thread_ID for retrieval. **5. Response Delivery** o Processed results (chatbot answers, summaries, quizzes, timestamps) are returned to the user as a structured response. o Database ensures fast reload for repeated queries. **6. Error Handling & Logging** o Any failed transcript loads or API errors are logged. o If LLM fails, fallback response is triggered with stored data from DB.
7	**Research Papers Published in the refereed/national/international journals in this duration (please attach copy): Yes/No** Yes
8	**Research Papers Published in Conference Proceedings/ Seminars in this duration(please attach copy) Yes/No** Yes
9	**Conferences/Seminars/workshops attended in this duration: Yes/No** Yes
10	**Any other achievements** No

Signature of students **Signature of Guide** **Signature of Project In charge**